ChatGPT & Gemini Prompt Engineering Master Reference Guide

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1. Foundational Architecture: How Generative Models Process Prompts

- Large Language Models predict next tokens based on probabilistic distribution over vocabulary.
- System Instructions set permanent behavioral guardrails, tone, and operational schema.
- Context Window Physics: Tokens in earlier positions receive slightly attenuated attention vs end tokens.
- Zero-Shot vs Few-Shot: Providing 2-3 exemplar input/output pairs dramatically increases schema adherence.

2. Battle-Tested System Instruction Templates

Role Definition: "You are a senior full-stack engineer and digital publishing architect."

Boundary Constraint: "Provide direct, production-ready code with no markdown preamble or placeholders."

Structured Output Request: "Always respond with strictly valid JSON adhering to the provided TypeScript interface."

```
interface PromptResponse { status: "success" | "error"; payload: Record<string, unknown>; }
```

3. Chain-of-Thought & Deterministic Parsing Strategies

- Instruct the model to reason through constraints step-by-step before emitting the final output.
- Enforce schema validation on server responses using Zod or JSON Schema before persisting to database.
- Fallback Handlers: Always catch JSON.parse errors and retry with temperature 0.0 for deterministic output.